

Executive Summary — WCDN Aegis Automation Protocol

The WCDN Aegis Automation Protocol demonstrates a fully automated cloud incident response pipeline designed to detect, triage, and contain security threats originating from a compromised EC2 instance. This project integrates AWS GuardDuty, Security Hub, EventBridge, CloudTrail, VPC Flow Logs, and a Lambda-based quarantine function to simulate real-world cloud security operations.

The environment included two EC2 instances, one serving as the compromised “Hydra-Infiltrator” node. GuardDuty successfully detected reconnaissance activity, suspicious outbound communication, SSH brute-force attempts, and privilege escalation behaviors. These findings were aggregated through Security Hub to provide centralized visibility and standardized alert formatting.

High-severity findings triggered an EventBridge rule configured to automatically invoke the JARVIS-Lockdown-Hydra Lambda function. This function isolated the affected instance by removing its IAM role, attaching an isolation security group, and tagging it for follow-up investigation. CloudWatch logs confirmed successful execution of all quarantine steps.

CloudTrail API logs and VPC Flow Logs provided corroborating evidence of unauthorized activity, including repeated failed authentication attempts, anomalous network traffic patterns, and outbound calls to known malicious IP addresses. These layers of visibility ensured high-confidence validation of detection accuracy.

The project demonstrated that automated detection and containment significantly reduce response time and improve overall cloud resilience. By integrating AWS-native services into a cohesive workflow, the Aegis Automation Protocol reflects industry-standard practices for SOC automation, threat detection engineering, and cloud incident response.

This project shows readiness for roles in cloud security engineering, SOC automation, detection engineering, and incident response, and forms a key component of the Wakanda Cyber Defense Network portfolio.